SECTION 26 32 13.13

GENERATOR SET – DIESEL FUELED

PART 1 GENERAL

1.01 SCOPE OF WORK

A. It is the intent of this specification to secure an engine driven generator set that has been prototype tested, factory built, production tested, and site tested, together with all accessories necessary for a complete installation as shown on the plans and drawings and specified herein.

B. The generator set shall be provided on the basis of a turbocharged diesel fueled engine generator set as further described in these specifications.

1.02 GENERAL REQUIREMENTS

A. The generator set will be a commercial design and will be complete with all of the necessary accessories for complete installation as shown on the plans, drawings, and specifications herein.

B. The equipment supplied and installed shall meet the requirements of the National Electrical Code, along with all applicable local codes and regulations.

C. All equipment shall be new and of current production of a national firm which manufactures the generator and assembles the generator sets as a complete and coordinated system. There will be one source responsibility for warranty, parts, and service through a local representative with factory-trained servicemen.

D. The generator must be no larger than 7’ x 3’.

1.03 SUBMITTAL

A. The submittal shall include prototype test certification and specification sheets showing all standard and optional accessories to be supplied, schematic wiring diagrams, dimension drawings, and interconnection diagrams identifying by terminal number, each required interconnection between the generator set, the transfer switch, and the SCADA system.

1.04 CODES AND STANDARDS

A. The generator set shall be listed to UL 2200.

B. The generator set shall conform to the requirements of the following codes and standards:

1. CSA C22.2, No. 14-M91 Industrial Control Equipment.
2. EN50082-2, Electromagnetic Compatibility-Generic Immunity Requirements, Part 2: Industrial.
3. EN55011, Limits and Methods of Measurement of Radio Interference Characteristics of Industrial, Scientific and Medical Equipment.
4. IEC8528 part 4, Control Systems for Generator Sets.
5. IEC Std 61000-2 and 61000-3 for susceptibility, 61000-6 radiated and conducted electromagnetic emissions.
6. IEEE446 Recommended Practice for Emergency and Standby Power Systems for Commercial and Industrial Applications.
7. NFPA 70, National Electrical Code, Equipment shall be suitable for use in systems in compliance to Article 700, 701, and 702.
8. NFPA 99, Essential Electrical Systems for Health Care Facilities.
9. NFPA 110, Emergency and Standby Power Systems. The generator set shall meet all requirements for Level 1 systems. Level 1 prototype tests required by this standard shall have been performed on a complete and functional unit. Component level type tests will not substitute for this requirement.

1.05 TESTING

A. To ensure that the equipment has been designed and built to the highest reliability and quality standards, the manufacturer and/or local representative shall be responsible for these separate tests: final production tests and site tests.

B. Final Production Tests: Each generator set shall be tested under varying loads with guards and exhaust system in place. Tests shall include:

1. Single-step load pickup.
2. Transient and steady-state governing.
3. Safety shutdown device testing.
4. Voltage regulation.
5. Rated Power @ 0.8 PF
6. Maximum Power.

Upon request a certified test record will be sent to the customer.

C. Site Tests: An installation check, start-up, and building load test shall be performed by the manufacturer's local representative. The engineer, regular operators, and the maintenance staff shall be notified of the time and date of the site test. The tests shall include:

1. Fuel, lubricating oil, and antifreeze shall be checked for conformity to the manufacturer's recommendations, under the environmental conditions present and expected.
2. Accessories that normally function while the set is standing by shall be checked prior to cranking the engine. These shall include block heaters, battery charger, generator strip heaters, remote annunciator, etc.
3. Start-up under test mode to check for exhaust leaks, path of exhaust gases outside the building, cooling air flow, movement during starting and stopping, vibration during running, normal and emergency line-to-line voltage and frequency, and phase rotation.
4. Automatic start-up by means of simulated power outage to test remote-automatic starting, transfer of the load, and automatic shutdown. Prior to this test, all transfer switch timers shall be adjusted for proper system coordination. Engine coolant temperature, oil pressure, and battery charge level along with generator voltage, amperes, and frequency shall be monitored throughout the test. An external load bank shall be connected to the system if sufficient station load is unavailable to load the generator to the nameplate kW rating.
5. Site load test shall be per NFPA 110 start-up and load-bank standards with a minimum of 4 hours with loading at 25%, 50%, 75%, and 100% load plus Single-step load pickup.

PART 2 PRODUCTS

2.01 EQUIPMENT

A. The generator shall be Stand-by rated at 15kW / 18.75kVA at 120/240 volts, 60 Hz., 0.80 power factor, 1 phase, 3-wire connected including radiator fan and all parasitic loads. It shall be sized to operate at the specified load at a maximum ambient of 104°F (40°C) and 1000 feet above sea level.

B. Motor starting performance and voltage dip determinations shall be based on the complete generator set. The generator set shall be capable of supplying a maximum of 47 skVA for starting motor loads with a maximum instantaneous voltage dip of 35%, as measured by a digital RMS transient recorder in accordance with IEEE standard 115. Motor starting performance and voltage dip determination that does not account for all components affecting total voltage dip i.e. engine, alternator, voltage regulator and governor will not be acceptable.

C. Engine brake horsepower shall be sufficient to deliver full rated generator set kW/kVA when operated at rated rpm and equipped with all engine-mounted parasitic and external loads such as radiator fan and power generator.

D. Vibration isolators shall be provided between the engine-generator and heavy-duty steel base.

E. The generator set shall be sound attenuated to a level suitable for a residential area.

F. The generator set shall be Cummins, Caterpillar, or Kohler.

2.02 ENGINE

A. The engine shall deliver the required electrical power at a governed speed of 1800 rpm. The engine shall be equipped with the following:

B. An electronic isochronous governor capable of +0.5% steady-state frequency regulation.

C. 12 Volt positive engagement solenoid shift-starting motor.

D. 70-Ampere minimum automatic battery charging alternator with solid-state voltage regulation.

E. Positive displacement, full pressure lubrication oil pump, cartridge oil filters, dipstick, and oil drain.

F. Dry-type replaceable air cleaner elements for normal applications.

G. Batteries shall be heavy duty SLI lead acid type with battery charger.

H. Engine-driven or electric fuel-transfer pump including fuel filter and electric solenoid fuel shutoff valve capable of lifting fuel.

I. The turbocharged, air-cooled engine shall be fueled by diesel.

J. The engine shall have a minimum of 4 cylinders, and be liquid-cooled by a unit-mounted radiator, blower fan, water pump, and thermostats. This system shall properly cool the engine in an ambient temperature up to 122°F (50°C).

K. The engine shall be EPA Tier 3 Stand-by certified from the factory, and shall not require a site performance test.

L. Engines shall start, achieve rated voltage, and freeway, and be capable of accepting load within 10 seconds of detected power interruption.

2.03 GENERATOR

A. The alternator shall be 4 pole, revolving field, brushless, 2/3-pitch, 12 lead, self-ventilated with drip-proof construction, amortisseur rotor windings, and skewed for smooth voltage waveform.

B. The ratings shall meet the NEMA standard (MG1-32.40) temperature rise limits. The insulation shall be class H per UL1446 and the varnish shall be a fungus resistant epoxy. Temperature rise of the rotor and stator shall be limited to 105°C. Alternator cooling shall be direct drive centrifugal blower.

C. The excitation system shall be of brushless PMG construction controlled by a solid-state voltage regulator capable of maintaining voltage within +/-0.5% at any constant load from 0% to 100% of rating.

D. The generator set shall meet the transient performance requirements of ISO 8528-5, level G-2.

E. The generator shall have a single maintenance-free bearing, shall be directly connected to the flywheel housing with a semi-flexible coupling between the rotor and the flywheel.

F. The generator shall be inherently capable of sustaining at least 250% of rated current for at least 10 seconds under a 3-phase symmetrical short circuit without the addition of separate current support devices.

2.04 CONTROLLER

A. Standards

1. The generator must meet NFPA-110 Level 1 requirements (1999 version) and must have an integral alarm horn as required by NFPA.
2. NFPA-99 and NEC must also be accommodated.
3. The controller shall be UL 508 listed.
4. Controller should be designed and produced by the generator manufacturer and must be overall coordinated with the genset. After market integrated controllers shall not be used.

B. Applicability

1. For standardization purposes, the control described herein must be available on generator sets 20 kW and larger.
2. The controller must be usable on 12- or 24-volt starting systems.
3. Environment
   i. -40 deg C to +70 deg C operating temperature range
   ii. 5-95% humidity, non-condensing

C. Hardware Requirements

1. The control shall have a run-off/reset-auto three-position selector switch.

2. A controller-mounted, latch-type emergency stop pushbutton.

3. Five indicating lights:
   i. System Ready
   ii. Not in AUTO
   iii. Programming Mode
   iv. System Warning
   v. System Shutdown

4. Display with two lines of 20 alphanumeric characters viewable in all light conditions.
5. Snap action sealed keypad for menu selection and data entry.
6. An audible alarm with alarm silence capability.
7. Panel lights shall be supplied as standard.

D. Control Functional Requirements

1. Field programmable time delay for engine start. Adjustment range, 0-15 minutes in 1 second increments.
2. Field programmable time delay engine cooldown. Adjustment range, 0-10 minutes in 1-minute increments.
3. Output for shedding loads if the generator reaches a user programmable percentage of its kW rating. Load shed must also be enabled if the generator output frequency falls below 59 Hz.
4. Programmable cyclic cranking that allows up to six crank cycles and up to 35 seconds of crank time per crank cycle.
5. Real-time clock and calendar for time stamping of events.
6. The control must be capable of detecting the following conditions and display on the control panel:
   i. Emergency stop
   ii. High coolant temperature
   iii. Controller internal fault
   iv. Low oil pressure
   v. Over-crank
   vi. Overspeed

7. Conditions resulting in generator warning (generator will continue to operate):
   i. Battery charger failure
   ii. Customer programmed digital auxiliary input
   iii. Customer programmed analog auxiliary input
   iv. Power system supplying load
   v. High battery voltage
   vi. High coolant temperature
   vii. Loss of AC sensing
   viii. Low battery voltage
   ix. Low coolant temperature

   x. Low fuel level or pressure

xi. Low oil pressure
xii. Overcurrent
xiii. Speed sensor fault
xiv. Weak battery

E. Control Monitoring Requirements

1. Controller interface must be an ethernet wired interface supporting open unencrypted communication with both TCP/IP using SNMP protocol and either Modbus TCP Standard or Modbus RTU Standard. Controllers must have with SNMP feature on board. Controller must not require proprietary software or interfaces to achieve remote communications for generator status monitoring and operational control (start/stop/etc)
2. All monitored functions must be viewable on the digital display.
3. The following generator functions must be monitored:
   i. All output voltages - single phase, three phase, line to line, and line to neutral, 1.0% accuracy
   ii. All single phase and three phase currents, 1.0% accuracy
   iii. Output frequency, 1.0 accuracy

4. Engine parameters listed below shall be monitored:
   i. Coolant temperature
   ii. Oil pressure
   iii. Battery voltage
   iv. RPM

5. Operational records shall be stored in the control beginning at system startup.
   i. Run time hours
   ii. Run time loaded hours
   iii. Run time unloaded hours
   iv. Number of starts
   v. Last run data including date, duration, and whether loaded or unloaded
   vi. Run time kilowatt hours

6. The following operational records shall be a resettable for maintenance purposes:
   i. Run time hours
   ii. Run time loaded hours
   iii. Run time unloaded hours
   iv. Number of starts
   v. Days of operation
   vi. Run time kilowatt hours

7. The controller shall store the last one hundred generator set system events with date and time of the event.

8. For maintenance and service purposes, the controller shall store and display on demand the following information:
   i. Manufacturer's model and serial number
   ii. Battery voltage
   iii. Generator set kilowatt rating
   iv. Rated current
   v. System voltage
   vi. System frequency
   vii. Number of phases

F. Accessories

1. PLC/SCADA Interface
   i. Generator set interface to the PLC/SCADA system hardwired I/O shall include:
      1. PLC digital input – Generator Run Status
      2. PLC digital input – Generator Fault
      3. PLC digital input – Generator Low Fuel

2. The generator shall come with a primary, factory installed, 70 ampere, 2-pole, 100% rated output circuit breaker that is UL listed. Load side breaker lugs shall be provided from the factory.
3. Engine block heater. Thermostatically controlled and sized to maintain manufacturers recommended engine coolant temperature to meet the start-up requirements of NFPA-99 and NFPA-110, Level 1.
4. Battery Charger. A 10-ampere automatic float to equalize battery charge. The battery charger shall be UL listed.
5. Battery rack, and battery cables, capable of holding the manufacturer's recommended batteries, shall be supplied.
6. Critical Silencer. The engine exhaust silencer shall be temperature and rust resistant, and rated for critical applications. The silencer will reduce total engine exhaust noise by 25-35 db(A).

2.05 GENERATOR WEATHER PROTECTIVE ENCLOSURE

A. The complete diesel engine generator set, including generator control panel, engine starting batteries and fuel oil tank, shall be enclosed in a factory assembled, weather protective, sound attenuated enclosure.

B. The enclosure must provide security from vandals and be aesthetically pleasing as well. In addition, the enclosure must meet applicable National Electric Code (NEC) and National Fire Protection Association (NFPA) codes relating to clearances of all items included with the Generator Set.

C. The performance of the enclosure must be in accordance with the Generator Set’s specific requirements for cooling and combustion airflow.

D. Clearances must be adequate for maintenance personnel and/or doors must be located such that service personnel have adequate access.

E. The enclosure will be of Modular construction allowing complete flexibility in design and use. The enclosure must be capable of being modified in the factory or after installation to meet various field conditions.

F. All enclosures are to be constructed from high strength, low alloy steel, aluminum or galvanized steel.

G. The enclosure shall be finish coated with powder baked paint for superior finish, durability, and appearance. Enclosures will be finished in the manufacturer's standard color.

H. The enclosure must be capable of withstanding a 100 mph wind. Intake louvers and enclosure overall must be able to maintain less the .01 ounces of moisture penetration per square foot of louver free area during a 4”/hr rainfall.

I. Lifting provisions must be provided in the base that enable the complete Gen-set with the enclosure to be lifted without damage.

J. The enclosure roof shall be pitched to prevent accumulation of water. The roof section shall be guttered.

K. The exhaust silencer shall be internally installed within the generator enclosure. Enclosure structure must be capable of supporting the largest commercially available silencer recommended for this particular model Gen-set. Exhaust silencer shall be mounted using heavy-duty 7 gage powder-coated brackets.

L. All doors shall be cross-braced to provide the maximum strength possible and to maintain alignment. Door seals are to be rubber material that is non-hydroscopic and will not allow the doors to freeze shut.

M. Doors must be hinged with stainless steel hinges and hardware and be removable.

N. Doors shall be equipped with lockable latches. Locks must be keyed alike.

O. The exhaust system shall include a roof penetration section that will eliminate rain or water run-off from entering the enclosure.

P. Engine crankcase emission canister, where required.

Q. The generator set shall be Level 2 sound attenuated. The generator set shall be provided with a sound attenuated enclosure. The generator set shall also include an exhaust silencing system and an air intake silencing system. The overall sound attenuating enclosure design shall reduce sound levels to a level acceptable in a residential area. The sound level shall be 71 dBA at 23 feet (7m) or less based on an (8) position average with a maximum position sound level not to exceed 73 dBA.

2.06 DOUBLE WALL SUB-BASE FUEL TANK

A. Provide a double wall sub-base tank constructed to meet all local codes and requirements. A fuel tank base of 24 hour capacity shall be provided as an integral part of the enclosure. It shall be contained in a rupture basin with 110% capacity. A locking fill cap, a mechanical reading fuel level gauge, low fuel level alarm contact, and fuel tank rupture alarm contact shall be provided.

1. The sub-base fuel system is listed under UL 142, sub section entitled Special Purpose Tanks EFVT category, and will bear their mark of UL Approval according to their particular classification.
2. The above ground steel secondary containment rectangular tank for use as a sub-base for diesel generators is manufactured and intended to be installed in accordance with the Flammable and Combustible Liquids Code—NFPA 30, the Standard for Installation and Use of Stationary Combustion Engine and Gas Turbines—NFPA 37, and Emergency and Standby Power Systems—NFPA 110.

B. Construction:

1. Primary Tank: It will be rectangular in shape and constructed in clam shell fashion to ensure maximum structural integrity and allow the use of a full throat fillet weld.
2. Steel Channel Support System: Reinforced steel box channel for generator support, with a load rating of 5,000 lbs. per gen set mounting hole location. Full height gussets at either end of channel and at gen set mounting holes shall be utilized.
3. Exterior Finish: The exterior coating has been tested to withstand continuous salt spray testing at 100 percent exposure for 244 hours to a 5 percent salt solution at 92-97 degrees F. The coating has been subjected to full exposure humidity testing to 100 percent humidity at 100 degrees F for 24 hours. Tests are to be conducted in accordance with The American Standard Testing Methods Society.

C. Venting:

1. Normal venting shall be sized in accordance with the American Petroleum Institute Standard No 2000, Venting Atmospheric and Low Pressure Storage Tanks not less than 1-1/4” (3 cm.) nominal inside diameter. A 1 - 1/4" atmospheric mushroom cap shall be furnished, piped, and installed above the highest fill point as a minimum

D. Emergency Venting

1. The emergency vent opening shall be sized to accommodate the total capacity of both normal and emergency venting and shall be not less than

that derived from NFPA 30, table 2-8, and based on the wetted surface area of the tank. The wetted area of the tank shall be calculated on the basis of 100 percent of the primary tank. A zinc plated emergency pressure relief vent cap shall be furnished for the primary tank. The vent is spring-pressure operated: opening pressure is 0.5/psig and full opening pressure is 2.5 psig. Limits are stamp marked on top of each vent. The emergency relief vent is sized to accommodate the total venting capacity of both normal and emergency vents.

E. Fuel Fill:

1. There shall be a 2" NPT opening within the primary tank with an 8" raised fill pipe and lockable manual fill cap.

F. Fuel Level:

1. A direct reading, UL listed, magnetic fuel level gauge with a hermetically-sealed, vacuum tested dial shall be provided to eliminate fogging.

G. Low Fuel Level Switch

1. Consists of a 50 watt float switch for remote or local annunciation of a (50% standard) low fuel level condition.

2.07 QUALITY ASSURANCE

A. The complete power generation system, including engine, generator, and switchgear, shall be the product of one manufacturer who has been regularly engaged in the production of complete generating systems for at least fifty (50) years. All components shall have been designed to achieve optimum physical and performance compatibility and prototype tested to prove integrated design capability. The complete system shall have been factory fabricated, assembled, and production tested. The naming of a specific manufacturer does not waive any requirements of this specification.

B. The generator engine and alternator shall not be manufactured by a Chinese company or their subsidiary.

C. Any and all exceptions to the published specifications shall be subject to the approval of the engineer.

2.08 RESPONSIBILITY

A. The responsibility for performance to this specification shall not be divided among individual component manufacturers, but must be assumed solely by the primary manufacturer. This includes generating system design, manufacture, test, and having a local supplier responsible for service, parts, and warranty for the total system.

2.09 MINIMUM SERVICE AND WARRANTY QUALIFICATIONS

A. The manufacturer shall have an authorized dealer facility within 2-hour drive time of the KUB service territory. This location shall have at least two generator technicians with full service trucks who can provide factory trained service for all generator components, including engines, alternators, and controllers. The facility shall have a full complement of required replacement parts, technical assistance, and warranty administration.

B. The generator technicians shall be available for service, 24 hours per day, 7 days per week, with a response time of 4 hours or less.

2.10 WARRANTY TERMS

A. The manufacturer's and dealer's extended warranty shall in no event be for a period of less than two (2) years from date of initial start-up of the system and shall include repair parts, labor, reasonable travel expense necessary for repairs at the jobsite, and expendables (lubricating oil, filters, antifreeze, and other service items made unusable by the defect) used during the course of repair. Applicable deductible costs shall be specified in the manufacturer's warranty. Running hours shall not be a limiting factor for the system warranty by either the manufacturer or servicing dealer. **Submittals received without written warranties as specified will be rejected in their entirety.**

2.11 DOCUMENTATION AND TRAINING

A. Furnish the services of a factory-authorized service representative to instruct owner's personnel in the operation, maintenance, and adjustment of the generator and related equipment. Provide a minimum of four (4) hours instruction scheduled seven (7) days in advance.

END OF SECTION

SECTION 26 32 13.13

GENERATOR SET – DIESEL FUELED

PART 1 GENERAL

1.01 SCOPE OF WORK

A. It is the intent of this specification to secure an engine driven generator set that has been prototype tested, factory built, production tested, and site tested, together with all accessories necessary for a complete installation as shown on the plans and drawings and specified herein.

B. The generator set shall be provided on the basis of a turbocharged diesel fueled engine generator set as further described in these specifications.

1.02 GENERAL REQUIREMENTS

A. The generator set will be a commercial design and will be complete with all of the necessary accessories for complete installation as shown on the plans, drawings, and specifications herein.

B. The equipment supplied and installed shall meet the requirements of the National Electrical Code, along with all applicable local codes and regulations.

C. All equipment shall be new and of current production of a national firm which manufactures the generator and assembles the generator sets as a complete and coordinated system. There will be one source responsibility for warranty, parts, and service through a local representative with factory-trained servicemen.

1.03 SUBMITTAL

A. The submittal shall include prototype test certification and specification sheets showing all standard and optional accessories to be supplied, schematic wiring diagrams, dimension drawings, and interconnection diagrams identifying by terminal number, each required interconnection between the generator set, the transfer switch, and the SCADA system.

1.04 CODES AND STANDARDS

A. The generator set shall be listed to UL 2200.

B. The generator set shall conform to the requirements of the following codes and standards:

1. CSA C22.2, No. 14-M91 Industrial Control Equipment.
2. EN50082-2, Electromagnetic Compatibility-Generic Immunity Requirements, Part 2: Industrial.
3. EN55011, Limits and Methods of Measurement of Radio Interference Characteristics of Industrial, Scientific and Medical Equipment.
4. IEC8528 part 4, Control Systems for Generator Sets.
5. IEC Std 61000-2 and 61000-3 for susceptibility, 61000-6 radiated and conducted electromagnetic emissions.
6. IEEE446 Recommended Practice for Emergency and Standby Power Systems for Commercial and Industrial Applications.
7. NFPA 70, National Electrical Code, Equipment shall be suitable for use in systems in compliance to Article 700, 701, and 702.
8. NFPA 99, Essential Electrical Systems for Health Care Facilities.
9. NFPA 110, Emergency and Standby Power Systems. The generator set shall meet all requirements for Level 1 systems. Level 1 prototype tests required by this standard shall have been performed on a complete and functional unit. Component level type tests will not substitute for this requirement.

1.05 TESTING

A. To ensure that the equipment has been designed and built to the highest reliability and quality standards, the manufacturer and/or local representative shall be responsible for these separate tests: final production tests and site tests.

B. Final Production Tests: Each generator set shall be tested under varying loads with guards and exhaust system in place. Tests shall include:

1. Single-step load pickup.
2. Transient and steady-state governing.
3. Safety shutdown device testing.
4. Voltage regulation.
5. Rated Power @ 0.8 PF
6. Maximum Power.

Upon request a certified test record will be sent to the customer.

C. Site Tests: An installation check, start-up, and building load test shall be performed by the manufacturer's local representative. The engineer, regular operators, and the maintenance staff shall be notified of the time and date of the site test. The tests shall include:

1. Fuel, lubricating oil, and antifreeze shall be checked for conformity to the manufacturer's recommendations, under the environmental conditions present and expected.
2. Accessories that normally function while the set is standing by shall be checked prior to cranking the engine. These shall include block heaters, battery charger, generator strip heaters, remote annunciator, etc.
3. Start-up under test mode to check for exhaust leaks, path of exhaust gases outside the building, cooling air flow, movement during starting and stopping, vibration during running, normal and emergency line-to-line voltage and frequency, and phase rotation.
4. Automatic start-up by means of simulated power outage to test remote-automatic starting, transfer of the load, and automatic shutdown. Prior to this test, all transfer switch timers shall be adjusted for proper system coordination. Engine coolant temperature, oil pressure, and battery charge level along with generator voltage, amperes, and frequency shall be monitored throughout the test. An external load bank shall be connected to the system if sufficient station load is unavailable to load the generator to the nameplate kW rating.
5. Site load test shall be per NFPA 110 start-up and load-bank standards with a minimum of 4 hours with loading at 25%, 50%, 75%, and 100% load plus Single-step load pickup.

PART 2 PRODUCTS

2.01 EQUIPMENT

A. The generator shall be Stand-by rated at 30kW / 37.5kVA at 120/240 volts, 60 Hz., 0.80 power factor, 1 phase, 3-wire connected including radiator fan and all parasitic loads. It shall be sized to operate at the specified load at a maximum ambient of 104°F (40°C) and 1000 feet above sea level.

B. Motor starting performance and voltage dip determinations shall be based on the complete generator set. The generator set shall be capable of supplying a maximum of 94 skVA for starting motor loads with a maximum instantaneous voltage dip of 35%, as measured by a digital RMS transient recorder in accordance with IEEE standard 115. Motor starting performance and voltage dip determination that does not account for all components affecting total voltage dip i.e. engine, alternator, voltage regulator and governor will not be acceptable.

C. Engine brake horsepower shall be sufficient to deliver full rated generator set kW/kVA when operated at rated rpm and equipped with all engine-mounted parasitic and external loads such as radiator fan and power generator.

D. Vibration isolators shall be provided between the engine-generator and heavy-duty steel base.

E. The generator set shall be sound attenuated to a level suitable for a residential area.

F. The generator set shall be Cummins, Caterpillar, or Kohler.

2.02 ENGINE

A. The engine shall deliver the required electrical power at a governed speed of 1800 rpm. The engine shall be equipped with the following:

B. An electronic isochronous governor capable of +0.5% steady-state frequency regulation.

C. 12 Volt positive engagement solenoid shift-starting motor.

D. 70-Ampere minimum automatic battery charging alternator with solid-state voltage regulation.

E. Positive displacement, full pressure lubrication oil pump, cartridge oil filters, dipstick, and oil drain.

F. Dry-type replaceable air cleaner elements for normal applications.

G. Batteries shall be heavy duty SLI lead acid type with battery charger.

H. Engine-driven or electric fuel-transfer pump including fuel filter and electric solenoid fuel shutoff valve capable of lifting fuel.

I. The turbocharged, air-cooled engine shall be fueled by diesel.

J. The engine shall have a minimum of 4 cylinders, and be liquid-cooled by a unit-mounted radiator, blower fan, water pump, and thermostats. This system shall properly cool the engine in an ambient temperature up to 122°F (50°C).

K. The engine shall be EPA Tier 3 Stand-by certified from the factory, and shall not require a site performance test.

L. Engines shall start, achieve rated voltage, and freeway, and be capable of accepting load within 10 seconds of detected power interruption.

2.03 GENERATOR

A. The alternator shall be 4 pole, revolving field, brushless, 2/3-pitch, 12 lead, self-ventilated with drip-proof construction, amortisseur rotor windings, and skewed for smooth voltage waveform.

B. The ratings shall meet the NEMA standard (MG1-32.40) temperature rise limits. The insulation shall be class H per UL1446 and the varnish shall be a fungus resistant epoxy. Temperature rise of the rotor and stator shall be limited to 105°C. Alternator cooling shall be direct drive centrifugal blower.

C. The excitation system shall be of brushless PMG construction controlled by a solid-state voltage regulator capable of maintaining voltage within +/-0.5% at any constant load from 0% to 100% of rating.

D. The generator set shall meet the transient performance requirements of ISO 8528-5, level G-2.

E. The generator shall have a single maintenance-free bearing, shall be directly connected to the flywheel housing with a semi-flexible coupling between the rotor and the flywheel.

F. The generator shall be inherently capable of sustaining at least 250% of rated current for at least 10 seconds under a 3-phase symmetrical short circuit without the addition of separate current support devices.

2.04 CONTROLLER

A. Standards

1. The generator must meet NFPA-110 Level 1 requirements (1999 version) and must have an integral alarm horn as required by NFPA.
2. NFPA-99 and NEC must also be accommodated.
3. The controller shall be UL 508 listed.
4. Controller should be designed and produced by the generator manufacturer and must be overall coordinated with the genset. After market integrated controllers shall not be used.

B. Applicability

1. For standardization purposes, the control described herein must be available on generator sets 20 kW and larger.
2. The controller must be usable on 12- or 24-volt starting systems.
3. Environment
   i. -40 deg C to +70 deg C operating temperature range
   ii. 5-95% humidity, non-condensing

C. Hardware Requirements

1. The control shall have a run-off/reset-auto three-position selector switch.

2. A controller-mounted, latch-type emergency stop pushbutton.

3. Five indicating lights:
    i. System Ready
    ii. Not in AUTO
    iii. Programming Mode
    iv. System Warning
    v. System Shutdown

4. Display with two lines of 20 alphanumeric characters viewable in all light conditions.
5. Snap action sealed keypad for menu selection and data entry.
6. An audible alarm with alarm silence capability.
7. Panel lights shall be supplied as standard.

D. Control Functional Requirements

1. Field programmable time delay for engine start. Adjustment range, 0-15 minutes in 1 second increments.
2. Field programmable time delay engine cooldown. Adjustment range, 0-10 minutes in 1-minute increments.
3. Output for shedding loads if the generator reaches a user programmable percentage of its kW rating. Load shed must also be enabled if the generator output frequency falls below 59 Hz.
4. Programmable cyclic cranking that allows up to six crank cycles and up to 35 seconds of crank time per crank cycle.
5. Real-time clock and calendar for time stamping of events.
6. The control must be capable of detecting the following conditions and display on the control panel:
    i. Emergency stop
    ii. High coolant temperature
    iii. Controller internal fault
    iv. Low oil pressure
    v. Over-crank
    vi. Overspeed

7. Conditions resulting in generator warning (generator will continue to operate):
    i. Battery charger failure
    ii. Customer programmed digital auxiliary input
    iii. Customer programmed analog auxiliary input
    iv. Power system supplying load
    v. High battery voltage
    vi. High coolant temperature
    vii. Loss of AC sensing
    viii. Low battery voltage
    ix. Low coolant temperature

    x. Low fuel level or pressure

xi. Low oil pressure
xii. Overcurrent
xiii. Speed sensor fault
xiv. Weak battery

E. Control Monitoring Requirements

1. Controller interface must be an ethernet wired interface supporting open unencrypted communication with both TCP/IP using SNMP protocol and either Modbus TCP Standard or Modbus RTU Standard. Controllers must have with SNMP feature on board. Controller must not require proprietary software or interfaces to achieve remote communications for generator status monitoring and operational control (start/stop/etc)
2. All monitored functions must be viewable on the digital display.
3. The following generator functions must be monitored:
   i. All output voltages - single phase, three phase, line to line, and line to neutral, 1.0% accuracy
   ii. All single phase and three phase currents, 1.0% accuracy
   iii. Output frequency, 1.0 accuracy

4. Engine parameters listed below shall be monitored:
   i. Coolant temperature
   ii. Oil pressure
   iii. Battery voltage
   iv. RPM

5. Operational records shall be stored in the control beginning at system startup.
   i. Run time hours
   ii. Run time loaded hours
   iii. Run time unloaded hours
   iv. Number of starts
   v. Last run data including date, duration, and whether loaded or unloaded
   vi. Run time kilowatt hours

6. The following operational records shall be a resettable for maintenance purposes:
   i. Run time hours
   ii. Run time loaded hours
   iii. Run time unloaded hours
   iv. Number of starts
   v. Days of operation
   vi. Run time kilowatt hours

7. The controller shall store the last one hundred generator set system events with date and time of the event.

8. For maintenance and service purposes, the controller shall store and display on demand the following information:
    i. Manufacturer's model and serial number
    ii. Battery voltage
    iii. Generator set kilowatt rating
    iv. Rated current
    v. System voltage
    vi. System frequency
    vii. Number of phases

F. Accessories

1. PLC/SCADA Interface
    i. Generator set interface to the PLC/SCADA system hardwired I/O shall include:
        1. PLC digital input – Generator Run Status
        2. PLC digital input – Generator Fault
        3. PLC digital input – Generator Low Fuel

2. The generator shall come with a primary, factory installed, 125 ampere, 2-pole, 100% rated output circuit breaker that is UL listed. Load side breaker lugs shall be provided from the factory.
3. Engine block heater. Thermostatically controlled and sized to maintain manufacturers recommended engine coolant temperature to meet the start-up requirements of NFPA-99 and NFPA-110, Level 1.
4. Battery Charger. A 10-ampere automatic float to equalize battery charge. The battery charger shall be UL listed.
5. Battery rack, and battery cables, capable of holding the manufacturer's recommended batteries, shall be supplied.
6. Critical Silencer. The engine exhaust silencer shall be temperature and rust resistant, and rated for critical applications. The silencer will reduce total engine exhaust noise by 25-35 db(A).

2.05 GENERATOR WEATHER PROTECTIVE ENCLOSURE

A. The complete diesel engine generator set, including generator control panel, engine starting batteries and fuel oil tank, shall be enclosed in a factory assembled, weather protective, sound attenuated enclosure.

B. The enclosure must provide security from vandals and be aesthetically pleasing as well. In addition, the enclosure must meet applicable National Electric Code (NEC) and National Fire Protection Association (NFPA) codes relating to clearances of all items included with the Generator Set.

C. The performance of the enclosure must be in accordance with the Generator Set's specific requirements for cooling and combustion airflow.

D. Clearances must be adequate for maintenance personnel and/or doors must be located such that service personnel have adequate access.

E. The enclosure will be of Modular construction allowing complete flexibility in design and use. The enclosure must be capable of being modified in the factory or after installation to meet various field conditions.

F. All enclosures are to be constructed from high strength, low alloy steel, aluminum or galvanized steel.

G. The enclosure shall be finish coated with powder baked paint for superior finish, durability, and appearance. Enclosures will be finished in the manufacturer's standard color.

H. The enclosure must be capable of withstanding a 100 mph wind. Intake louvers and enclosure overall must be able to maintain less the .01 ounces of moisture penetration per square foot of louver free area during a 4"/hr rainfall.

I. Lifting provisions must be provided in the base that enable the complete Gen-set with the enclosure to be lifted without damage.

J. The enclosure roof shall be pitched to prevent accumulation of water. The roof section shall be guttered.

K. The exhaust silencer shall be internally installed within the generator enclosure. Enclosure structure must be capable of supporting the largest commercially available silencer recommended for this particular model Gen-set. Exhaust silencer shall be mounted using heavy-duty 7 gage powder-coated brackets.

L. All doors shall be cross-braced to provide the maximum strength possible and to maintain alignment. Door seals are to be rubber material that is non-hydroscopic and will not allow the doors to freeze shut.

M. Doors must be hinged with stainless steel hinges and hardware and be removable.

N. Doors shall be equipped with lockable latches. Locks must be keyed alike.

O. The exhaust system shall include a roof penetration section that will eliminate rain or water run-off from entering the enclosure.

P. Engine crankcase emission canister, where required.

Q. The generator set shall be Level 2 sound attenuated. The generator set shall be provided with a sound attenuated enclosure. The generator set shall also include an exhaust silencing system and an air intake silencing system. The overall sound attenuating enclosure design shall reduce sound levels to a level acceptable in a residential area. The sound level shall be 71 dBA at 23 feet (7m) or less based on an (8) position average with a maximum position sound level not to exceed 73 dBA.

2.06 DOUBLE WALL SUB-BASE FUEL TANK

A. Provide a double wall sub-base tank constructed to meet all local codes and requirements. A fuel tank base of 24 hour capacity shall be provided as an integral part of the enclosure. It shall be contained in a rupture basin with 110% capacity. A locking fill cap, a mechanical reading fuel level gauge, low fuel level alarm contact, and fuel tank rupture alarm contact shall be provided.

1. The sub-base fuel system is listed under UL 142, sub section entitled Special Purpose Tanks EFVT category, and will bear their mark of UL Approval according to their particular classification.
2. The above ground steel secondary containment rectangular tank for use as a sub-base for diesel generators is manufactured and intended to be installed in accordance with the Flammable and Combustible Liquids Code—NFPA 30, the Standard for Installation and Use of Stationary Combustion Engine and Gas Turbines—NFPA 37, and Emergency and Standby Power Systems—NFPA 110.

B. Construction:

1. Primary Tank: It will be rectangular in shape and constructed in clam shell fashion to ensure maximum structural integrity and allow the use of a full throat fillet weld.
2. Steel Channel Support System: Reinforced steel box channel for generator support, with a load rating of 5,000 lbs. per gen set mounting hole location. Full height gussets at either end of channel and at gen set mounting holes shall be utilized.
3. Exterior Finish: The exterior coating has been tested to withstand continuous salt spray testing at 100 percent exposure for 244 hours to a 5 percent salt solution at 92-97 degrees F. The coating has been subjected to full exposure humidity testing to 100 percent humidity at 100 degrees F for 24 hours. Tests are to be conducted in accordance with The American Standard Testing Methods Society.

C. Venting:

1. Normal venting shall be sized in accordance with the American Petroleum Institute Standard No 2000, Venting Atmospheric and Low Pressure Storage Tanks not less than 1-1/4” (3 cm.) nominal inside diameter. A 1 - 1/4" atmospheric mushroom cap shall be furnished, piped, and installed above the highest fill point as a minimum

D. Emergency Venting

1. The emergency vent opening shall be sized to accommodate the total capacity of both normal and emergency venting and shall be not less than

that derived from NFPA 30, table 2-8, and based on the wetted surface area of the tank. The wetted area of the tank shall be calculated on the basis of 100 percent of the primary tank. A zinc plated emergency pressure relief vent cap shall be furnished for the primary tank. The vent is spring-pressure operated: opening pressure is 0.5/psig and full opening pressure is 2.5 psig. Limits are stamp marked on top of each vent. The emergency relief vent is sized to accommodate the total venting capacity of both normal and emergency vents.

E. Fuel Fill:

1. There shall be a 2" NPT opening within the primary tank with an 8" raised fill pipe and lockable manual fill cap.

F. Fuel Level:

1. A direct reading, UL listed, magnetic fuel level gauge with a hermetically-sealed, vacuum tested dial shall be provided to eliminate fogging.

G. Low Fuel Level Switch

1. Consists of a 50 watt float switch for remote or local annunciation of a (50% standard) low fuel level condition.

2.07 QUALITY ASSURANCE

A. The complete power generation system, including engine, generator, and switchgear, shall be the product of one manufacturer who has been regularly engaged in the production of complete generating systems for at least fifty (50) years. All components shall have been designed to achieve optimum physical and performance compatibility and prototype tested to prove integrated design capability. The complete system shall have been factory fabricated, assembled, and production tested. The naming of a specific manufacturer does not waive any requirements of this specification.

B. The generator engine and alternator shall not be manufactured by a Chinese company or their subsidiary.

C. Any and all exceptions to the published specifications shall be subject to the approval of the engineer.

2.08 RESPONSIBILITY

A. The responsibility for performance to this specification shall not be divided among individual component manufacturers, but must be assumed solely by the primary manufacturer. This includes generating system design, manufacture, test, and having a local supplier responsible for service, parts, and warranty for the total system.

2.09 MINIMUM SERVICE AND WARRANTY QUALIFICATIONS

A. The manufacturer shall have an authorized dealer facility within 2-hour drive time of the KUB service territory. This location shall have at least two generator technicians with full service trucks who can provide factory trained service for all generator components, including engines, alternators, and controllers. The facility shall have a full complement of required replacement parts, technical assistance, and warranty administration.

B. The generator technicians shall be available for service, 24 hours per day, 7 days per week, with a response time of 4 hours or less.

2.10 WARRANTY TERMS

A. The manufacturer's and dealer's extended warranty shall in no event be for a period of less than two (2) years from date of initial start-up of the system and shall include repair parts, labor, reasonable travel expense necessary for repairs at the jobsite, and expendables (lubricating oil, filters, antifreeze, and other service items made unusable by the defect) used during the course of repair. Applicable deductible costs shall be specified in the manufacturer's warranty. Running hours shall not be a limiting factor for the system warranty by either the manufacturer or servicing dealer. **Submittals received without written warranties as specified will be rejected in their entirety.**

2.11 DOCUMENTATION AND TRAINING

A. Furnish the services of a factory-authorized service representative to instruct owner's personnel in the operation, maintenance, and adjustment of the generator and related equipment. Provide a minimum of four (4) hours instruction scheduled seven (7) days in advance.

END OF SECTION

Bid Terms and Conditions

1. Bidders must submit pricing for all items requested. Unit prices will prevail.
2. Bidder must complete the bid package in its entirety and provide all information requested in the bid package.
3. Bidder must acknowledge any Addendums (if applicable) and include with bid package.
4. New Bidders will be required to provide a completed W-9 and Vendor Application prior to issuance of Purchase Order.
5. Bids are valid and may not be withdrawn until 60 days after the bid due date. Should there be reasons why the Bid cannot be awarded within the specified period, the time may be extended by mutual agreement between the Owner and the Bidder(s).
6. All services, goods, and/or equipment are FOB Destination.
7. Bidder hereby acknowledges that the services, goods, and/or equipment will be provided in strict accordance with the KUB bid specifications.
8. Failure to comply with any of the Bid Terms and Conditions may result in a bid rejection. KUB reserves the right to reject any or all bids at KUB's sole discretion.
9. Purchase Orders will only be issued to awarded Bidder that submitted bid. No Purchase Order will be issued to third party vendors.

By signing below, Bidder acknowledges and accepts the above Bid Terms and Conditions and attached Terms of Purchase Order (Exhibit A).

Date:

Signed:

Title:

Company:

MBE (Minority Business Enterprise) / WBE (Women Business Enterprise) ☐ Yes ☐ No

Does your business qualify as a KUB-defined Small Business Enterprise (fifty or fewer employees and less than $10 million in annual revenue)? ☐ Yes ☐ No

EXHIBIT A

Terms of Purchase Order

The Knoxville Utilities Board ("KUB"), a municipal utility created and existing pursuant to the Charter of the City of Knoxville, a municipality existing under the laws of the State of Tennessee; hereby orders the services, goods and/or equipment on the front of this purchase order ("P.O."), subject to and upon the express terms and conditions printed and written on the front and reverse hereof. All invoices, labels, shipping documents and correspondence must contain the purchase order number printed on the form hereof.

1. Acceptance-Vendor(s) acceptance of this order is expressly to the terms and conditions contained herein and the Bid Terms and Conditions, if applicable, and acceptance with any additional conditions or modifications to this P.O. will be deemed a rejection of this order. If no notification is received by KUB within 15 days after vendor's receipt of the P.O. or if shipment of any good and/or equipment of any part is made, then it shall constitute an acceptance by the vendor(s) of the terms and conditions of this P.O.
2. Performance-Performance of this P.O. must be in accordance with its terms, dating, and conditions. No variation or modification of this P.O. is binding upon KUB unless such variation or modification is agreed to in writing by an authorized representative of the Procurement Department of KUB.
3. Delivery-Time of delivery at KUB's designated receiving facility is of the essence with respect to this P.O. If goods, equipment, or services delivery fail in any respect to conform to the terms of this P.O. or are otherwise non-conforming, KUB may in its sole discretion (1) reject the whole or (2) accept the whole or (3) accept any part and reject the balance or (4) cancel the order and order the equipment, goods, and/or services from another source and recover from the seller any additional costs incurred in ordering from another source and any other remedies under the Uniform Commercial Code codified at T.C.A. §§47-2-101 et. seq.
4. Price-Unless otherwise stated in this P.O., the price for the goods, equipment, and/or services covered hereby shall be the price quoted or contract price to KUB, which shall include and account for all Federal, State, and local fees, duties, tariffs, and taxes that are applicable at the time of contracting or that may become applicable at any time thereafter. The price quoted or contract price to KUB is a fixed price that is not subject to change by reason of increased or newly imposed Federal, State, and local fees, duties, tariffs, and taxes that have not been included and accounted for in Vendor's bid price and become applicable after the time of contracting. Any allowable discount for prompt payment is to be calculated from the date the invoice is received in proper form or from the date the goods or equipment is received whichever date is later. Payment of invoices for major equipment, which is ordinarily tested or inspected for conformity to specifications before being put into service, will be held until certain testing or verification is completed, but not in excess of sixty (60) days from receipt of equipment, unless equipment is found to be defective or non-conforming.
5. Payment – Payment terms will be NET 30 DAYS from the date of approval by KUB of Vendor(s) invoice.
6. Count-KUB's count of goods and materials shall be accepted as conclusive on all shipments. Charges for extras including but not limited to packing, loading, drayage, dunnage, or cartage shall not be accepted unless specifically stated on this P.O. No charges will be accepted for packaging, boxes, drums, barrels, reels, cores, etc.
7. Amendments-No agreement or understanding with respect to the amendment or modification of this P.O. shall be binding upon KUB unless approved in writing by an authorized representative of the Procurement Department of KUB. Each transaction between KUB and vendor is separate and distinct. No waiver of any breach of any terms or condition of this P.O. shall be construed as a waiver of any subsequent breach of that term or condition of the same or different nature of this or any other P.O. or contract of KUB related to the goods, equipment, and/or services ordered under this P.O.
8. Compliance-By accepting this P.O. Vendor represents, warrants, and guarantees that all applicable provisions of federal, state, and local laws, ordinances, codes, rules, and regulations which are applicable to the manufacture and/or sale of the goods, services, and/or equipment have been complied with by the Vendor. Vendor agrees to indemnify, hold harmless and defend KUB, its commissioners, officers, managers, employees and agents from and against all demands, claims, liabilities, losses, damages, fines and the costs and expenses incident thereto, including reasonable attorney's fees, made against KUB, or which KUB may become responsible for, as a result of death or bodily injury to any person, damage or destruction to property, the violation of any law, regulation, or order or any other claims of loss or damage against KUB, by any third party, arising out of, resulting from, or related to the obligations of Vendor under this Purchase Order, Vendor's breach of any term or provision of this Purchase Order or because of any negligent or willful act or omission of Vendor, its employees, agents or subcontractors. This agreement to indemnify, hold harmless and defend KUB shall include claims related to KUB's own negligence, except that it shall not apply to losses and/or liabilities arising from the sole negligence of KUB. KUB and Vendor agree that this provision of indemnity does not in any way constitute a waiver by KUB of its protections under the Governmental Tort Liability Act as codified in T.C.A. §§29-20-101 et. seq.
9. Nondiscrimination-KUB is an equal opportunity employer and as such requires that its vendors not discriminate on the basis of race, color, sex, religion, or ethnic origin. Acceptance of this P.O. constitutes vendor's acknowledgement of this provision. KUB encourages the use of small-business, minority, and women-owned business enterprises.
10. Choice of Law-This order and any subsequent contract related to this P.O. shall be governed by and construed with the laws of the State of Tennessee.
11. Force Majeure-Upon any discontinuance or substantial interference with KUB's business by reason of fire, flood, earthquake, and other acts of God, embargo, civil disturbance, acts of terrorism, governmental regulation, including, but not limited to, government-imposed fees, duties, tariffs, and taxes, or causes beyond KUB's control, KUB shall have the option of canceling any unfilled portion of this P.O. upon reasonable notice to vendor.
12. Default – KUB may by written notice of default to Vendor cancel the whole or part of this P.O. or exercise any other remedy provided to KUB by law or in equity including any remedy under the Uniform Commercial code as codified at T.C.A. §§47-1-101 et. seq. in any of the following circumstances 1) Vendor fails to make delivery of the goods within the time specified, 2) Vendor is in breach of any of the terms or conditions of the P.O. or 3) Vendor becomes insolvent or makes assignment for the benefit of creditors, or if there shall be instituted by or against Vendor any proceedings under any bankruptcy, reorganization, arrangement, readjustment or debt or insolvency law of any jurisdiction or for the appointment of a receiver or trustee in respect to any of vendor's property.
13. Title VI – The Vendor, Supplier, or Contractor agrees that it will comply with Title VI of the Civil Rights Act of 1964 and all requirements imposed by KUB to the effect that no person shall, on the grounds of race, color, or national origin, be excluded from participating in, or denied the benefits of, or be subject to discrimination under any program or activities for which KUB received Federal Financial Assistance. The Vendor, Supplier, or Contractor agrees to compile appropriate data, maintain records, and submit reports as required to permit effective enforcement of Title VI. If there are any violations of this assurance, KUB shall have the right to recommend corrective actions or to seek administrative enforcement of this assurance, up to and including termination of federal contracts involving the receipt of financial assistance.
14. Each of the parties to this Agreement hereby certifies to the other party or parties that, pursuant to Tennessee Code Annotated Section 12-4-119, such party is not currently and will not for the duration of this Agreement engage in a boycott of Israel. For purposes of this section, all definitions, terms, and meanings in Tennessee Code Annotated Section 12-4-119 shall be incorporated herein by reference to give full force and effect to the intent of the statute.

Evaluated Bid: *Generator Replacement*, Control #*4053*

I. **Statement of Intent** - Knoxville Utilities Board (KUB) is seeking bids from Suppliers for KUB's ***Generator Replacement*.** Proposed bids should meet specifications. Please provide outright purchase price as per unit cost.

II. **Overview of KUB** – KUB is a governmental agency existing under the laws of Tennessee and an independent agency of the City of Knoxville. KUB provides electricity, fiber, gas, water, and wastewater services to more than 525,000 customers in Knoxville and parts of seven surrounding counties. The KUB electric system is one of the nation's most dependable, providing uninterruptible service 99.98% of the time.

III. **Request for Bid (RFB) Schedule**:

RFB Issue Date	February 11th, 2026
Cutoff for Questions	February 20th, 2026, at 5:00 PM, EST
Bids due no later than	**February 26th, 2026, at 2pm, EST**
Award on or around	March 9th, 2026

IV. **Evaluated Bid Requirements** –

A. Read the RFB in its entirety.
B. Submit a complete and detailed Bid through KUB's online bid portal, **www.BidNetDirect.com.**
C. Submit all questions in writing no later than February 20th, 2026, at 5:00 PM, EST (by email to Tanner.Hawkins@kub.org or via www.BidNetDirect.com).
D. Bidder must acknowledge and accept the KUB's Terms and Conditions.
E. Bidder must include Attachment A – Price Sheet with any applicable information.
F. Bidder must include Attachment B – References.
G. Bidder must include Attachment C – Contact Information.
H. Bidder must respond to this bid with strict adherence to the Spec Sheet(s).
I. All line items are FOB destination (include shipping and freight charges with unit prices). Include all lead time information in your submission.
J. If Bidder has never done business with KUB, attach a copy of a completed W-9 (Request for Taxpayer Identification Number and Certification) or Social Security number for 1099 Reporting.
K. Does the Bidder qualify as an MBE (Minority Business Enterprise), WBE (Women Owned Business Enterprise), or SBE (Small Business Enterprise)? If so, provide proof from a certifying organization you are certified.

V. **Evaluation Criteria** – An evaluation team consisting of KUB representatives will review and independently rate each bid in the following order of importance:

A. Price – Attachment A Price Sheet.

B. Firm Description and Adherence to Functional Requirements/Spec.

C. Dealer Performance/Experience/References – Attachment B References.

Attachment A

PRICING SHEET

[Insert price breakdown descriptions]

Attachment B

REFERENCE LISTING

List reference(s) that most closely reflect similar contracts within the past five (5) years. Copy this page to provide a separate "Reference Listing" for each Reference. Please also include examples of previous work on similar projects (i.e. Pictures).

Name of Company, Utility, City, etc.

____

Address:

____

Contact Name: __________________________ Title: __________________________________

Phone: _______________________________ Email: __________________________________

Service Dates: ____

Examples of previous work: ____

Name of Company, Utility, City, etc.

____

Address:

____

Contact Name: __________________________ Title: __________________________________

Phone: _______________________________ Email: __________________________________

Service Dates: ____

Examples of previous work: ____

Attachment C

CONTACT INFORMATION

Company Name: ____

Company Address: ____

____

____

Contact Name: ____

Title: ____

Phone: ______________________ Fax: _______________________ Mobile: _______________________

Email: ____

Contact Name: ____

Title: ____

Phone: ______________________ Fax: _______________________ Mobile: _______________________

Email: ____

Contact Name: ____

Title: ____

Phone: ______________________ Fax: _______________________ Mobile: _______________________

Email: ____

Contact Name: ____

Title: ____

Phone: ______________________ Fax: _______________________ Mobile: _______________________

Email: ____